

B. Tech. Semester - 3rd
Mid Term Examination
Course Code – MC 209
Duration 90 Minutes

Roll No. - 23/MC/65..

Sept. 2024

Database Management System

M.M. – 20

Note : All questions are compulsory. Assume suitable missing data, if any.

Q1.

(C04) (6)

Consider the relation schema $R(A, B, C, D, E, F)$ with functional dependencies $AC \rightarrow B$, $BD \rightarrow F$ and $F \rightarrow CE$.

1. List all the candidate keys of R .
2. Is R in 3NF? If yes, justify. If no, specify at least one FD which violates the definition?
3. Which FD(s) (if any) of R violates BCNF?
4. Consider the decomposition of R into $R_1(A, B, C)$, $R_2(C, E, F)$ and $R_3(A, D, F)$. Give YES/NO answers for the following:
 - i. Is this decomposition lossless?
 - ii. Is this decomposition dependency preserving?
 - iii. Is this decomposition in BCNF (i.e. are R_1 , R_2 and R_3 all in BCNF)?

Q2.

(C04) (1)

Define the Multivalued Functional Dependencies with the help of an example.

Q3.

(C04) (2)

Consider the following Schema and Functional Dependencies – $T(A, B, C, D, E, F)$
 $FD = \{A \rightarrow BC, AC \rightarrow DEF, F \rightarrow AB\}$. Give the minimal cover of the set of functional dependencies.

Q4.

(C02) (3)

Construct an E-R diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents.

Q5.

(C01) (3)

Define redundancy with respect to data storage. How can the database approach help in controlling redundancy?

Q6.

(C01) (5)

Define the ER diagram for the following problem "Airline Database". The data held by the following tables do not exist independently. Hence, there are a number of inter-relationships that must be considered. Also, sketch the corresponding Database Schema. The relationships between the tables in the Airline Database are as follows: